

Arc length of an elliptical helix via Ramanujan's second perimeter approximation, with explicit pitch thresholds

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Abstract

We observe that one turn of the elliptical helix $\mathbf{r}(t) = (a \cos t, b \sin t, ct)$ has exactly the length of the ellipse with semi-axes $\sqrt{a^2 + c^2}$ and $\sqrt{b^2 + c^2}$. Consequently Ramanujan's second approximation for the perimeter of an ellipse gives a closed-form turn length, and the pitch makes the approximation more accurate. We give explicit thresholds: if $c \geq 0.197254a$ the relative error is below 10^{-6} for every $b \in [0, a]$; if $c \geq 0.534032a$ it is below 10^{-9} . The result gives integral-free wire and path lengths for elliptical coil springs, elliptical helical antennas and helical machining paths. Version 2 adds an exact formula for coils with stepped pitch and part turns, and closed-form estimates for progressive-pitch and tapered elliptical coils.

1 The reduction

Let $a \geq b \geq 0$, $a > 0$, $c > 0$, and write $A = \sqrt{a^2 + c^2}$, $B = \sqrt{b^2 + c^2}$.

Lemma 1. *The length of one turn of $\mathbf{r}(t) = (a \cos t, b \sin t, ct)$, $0 \leq t \leq 2\pi$, equals the perimeter $P(A, B)$ of an ellipse with semi-axes A and B .*

Proof. $|\mathbf{r}'(t)|^2 = a^2 \sin^2 t + b^2 \cos^2 t + c^2 = (a^2 + c^2) \sin^2 t + (b^2 + c^2) \cos^2 t = A^2 \sin^2 t + B^2 \cos^2 t$, which is the squared speed of $(A \cos t, B \sin t)$. Integrating over $[0, 2\pi]$ gives $P(A, B) = 4A E(1 - B^2/A^2)$. $\square$

2 Error of Ramanujan's approximation

Ramanujan's second approximation [1] is

$$R(A, B) = \pi(A + B) \left(1 + \frac{3h}{10 + \sqrt{4 - 3h}} \right), \quad h = \left(\frac{A - B}{A + B} \right)^2.$$

Since $P/(\pi(A+B)) = \sum_{n \geq 0} \binom{1/2}{n}^2 h^n$, the difference $(P - R)/(\pi(A+B))$ has vanishing coefficients up to h^4 and leading term $\frac{3}{2^{17}} h^5$; we verified symbolically that its coefficients are positive through h^{25} , and on a 2001-point grid (50-digit arithmetic) that the relative error $\varepsilon(h) = (P - R)/P$ is increasing. This agrees with the error analyses in [2, 3].

Because $b \geq 0$, $B/A \geq k(\rho) := \rho/\sqrt{1 + \rho^2}$ with $\rho = c/a$, hence $h \leq h_{\max}(\rho) = ((1 - k)/(1 + k))^2$.

Theorem 1. *For the helix above, the relative error of $R(A, B)$ as an approximation of the turn length satisfies $\varepsilon \leq \varepsilon(h_{\max}(c/a))$, with equality at $b = 0$. In particular*

$$\frac{c}{a} \geq 0.197254 \implies \varepsilon < 10^{-6} \quad \text{for every } b \in [0, a].$$

At $c = 0.2a$ one gets $k = 0.196116\dots$, $h_{\max} = 0.451689\dots$ and worst-case error 9.3758×10^{-7} .

worst-case relative error $\leq$	required pitch $c/a \geq$
10^{-4}	0.0314394
10^{-5}	0.1057625
10^{-6}	0.197254
10^{-7}	0.300449
10^{-8}	0.4129297
10^{-9}	0.5340313

Table 1: Pitch thresholds (worst case over $b \in [0, a]$). For $c/a \rightarrow 0$ the worst case tends to the classical 4.02×10^{-4} of the flat ellipse.

3 Numerical check

For 300 random helices with $a = 1$, $b \in (0, 1)$, $c \in (0.2, 3)$ the turn length was computed by direct adaptive quadrature of $|\mathbf{r}'(t)|$ and compared with $R(A, B)$; the largest relative deviation was 8.17×10^{-8} , consistent with the theorem.

4 Applications

The formula gives the wire length per turn of elliptical coil springs and elliptical helical antennas (studied since [4]) and the path length of helical machining or in-pipe robot trajectories, without evaluating an elliptic integral, at part-per-million accuracy whenever $c \geq 0.2a$.

5 Extensions (added in version 2)

Throughout, a, b are the semi-axes of the winding, $c_i = p_i/2\pi$, $A_i = \sqrt{a^2 + c_i^2}$, $B_i = \sqrt{b^2 + c_i^2}$, and all lengths were compared with direct quadrature of $|\mathbf{r}'(t)|$ at 30 digits.

Stepped pitch and part turns (exact). A coil made of sections of n_i turns at constant pitch p_i has length $\sum_i n_i P(A_i, B_i)$ when every n_i is an integer. A part turn is an incomplete elliptic integral: with $m = 1 - B^2/A^2$,

$$\int_0^\varphi \sqrt{A^2 \sin^2 t + B^2 \cos^2 t} dt = A[E(m) - E(\frac{\pi}{2} - \varphi, m)] \quad (0 \leq \varphi \leq \frac{\pi}{2}),$$

and each further quarter turn adds $A E(m)$; if the part turn ends φ' radians into an odd-numbered quarter (one starting at $t = \pi/2 + k\pi$), the last piece is $A E(\varphi', m)$ instead. For the sections (1.5, 3), (6, 15), (1.25, 4) (turns, pitch in mm) on $a = 10$, $b = 6$ this gives 460.37961606241 mm, equal to the quadrature value to 30 digits. Replacing P by Ramanujan's R gives integral-free lengths with the error of Theorem 1.

Progressive pitch. For pitch rising linearly from p_0 to p_1 over N turns, Simpson's rule applied turn by turn to $R(\sqrt{a^2 + c^2}, \sqrt{b^2 + c^2})$, with c at the start, middle and end of each turn, gave relative errors 1.2×10^{-6} (p : 4 $\rightarrow$ 16 mm, $N = 8$), 5.8×10^{-6} ($5 \rightarrow 30$, $N = 6$) and 2.0×10^{-8} ($10 \rightarrow 12$, $N = 10$).

Tapered coils. For a round conical helix $(r(t) \cos t, r(t) \sin t, ct)$ with $r(t) = r_0 + \alpha t$ the length is the classical closed form $[r\sqrt{r^2 + k^2} + k^2 \operatorname{arsinh}(r/k)]/(2\alpha)$ between r_0 and r_1 , $k^2 = \alpha^2 + c^2$. For an elliptical taper $a(t) = a_0 + \alpha t$, $b(t) = b_0 + \beta t$, summing $R(\sqrt{a^2 + \beta^2 + c^2}, \sqrt{b^2 + \alpha^2 + c^2})$ at mid-turn values gave relative errors 8.3×10^{-6} , 1.9×10^{-6} and 1.0×10^{-7} on three test coils.

Changes in version 2. Section 5 is new. Two threshold entries of Table 1 were rounded up so that each stated threshold is sufficient (version 1 gave 0.300448 and 0.4129296).

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References

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